

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

Make up Examinations - May - June 2023

21BT302 – BIOCHEMISTRY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Differentiate between homo and heteropolysaccharides with examples. Add a note on glycosaminoglycans.	10	L*2	1	1.2.1
	b) Discuss the properties of lipids. List its functions.	10	L2	2	1.2.1
2.	a) With examples, explain classification of carbohydrates.	10	L2	1	1.2.1
	b) What are sphingolipids? Explain its structural details.	10	L2	2	1.2.1
3.	a) Depicting the structure, explain how amylopectin and cellulose differs in their structural features. Add a note on its function.	10	L2	1	1.2.1
	b) Give an account on classification of lipids.	10	L2	2	1.2.1
Unit – II					
4.	a) What are peptides? Add a note on tertiary and quaternary structure of proteins.	10	L2	3	1.2.1
	b) With a neat diagram, explain the secondary structure of t-RNA.	10	L2	4	1.2.1
5.	a) Classify proteins and add a note on globular proteins.	10	L2	3	1.2.1
	b) What are nucleosomes? Discuss different levels of complex arrangements of DNA during chromatin organization.	10	L2	4	1.2.1
6.	a) Give an account on classification of amino acids.	10	L2	3	1.2.1
	b) Explain different molecular models of DNA.	10	L2	4	1.2.1
Unit – III					
7.	a) Give an account on beta oxidation of fatty acids.	10	L2	5	1.2.1
	b) Explain pentose phosphate pathway. Add a note on its importance.	10	L2	5	1.2.1
8.	a) Depict the process of glycolysis with its significance.	10	L2	5	1.2.1
	b) Illustrate TCA cycle. Why is it called amphibolic pathway? Explain.	10	L2	5	1.2.1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
